Sim-to-Real Joint Tracking with PPO

Can varied training physics improve performance in an unfamiliar simulation?

Mechatronics engineering student demonstration



1 joint • 2 policies

120 evaluation episodes

SAY “My question is whether varied training physics helps a controller handle an unfamiliar simulated joint.”

Why this is a mechatronics problem

A robot controller meets more than an ideal equation.

MECHANICS

Inertia changes acceleration.

Friction changes damping.

ACTUATION

Delayed commands arrive late.

Torque has physical limits.

SENSING

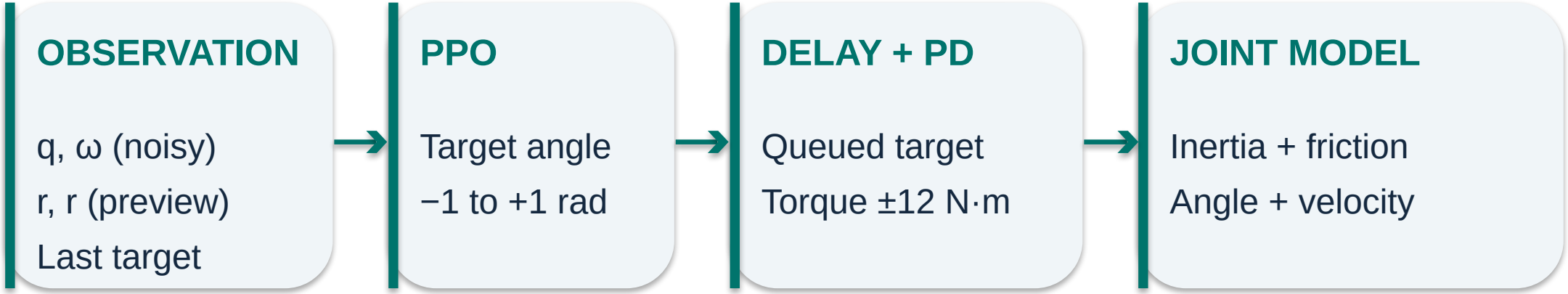
Measurements contain noise.

Feedback quality affects control.

Inspired by Go2 / ANYmal applications; this model isolates one joint.

SAY “A useful controller must work with the motor, the sensors, and the mechanical system together.”

PPO chooses a target; PD supplies torque



Noisy state returns to both the policy and PD feedback.

$$\tau = \text{clip}[12(q_{target,\,delayed} - q_{measured}) - 2\omega_{measured},\ -12,\ 12]$$

Both policy and PD updates run at 20 Hz in this simplified implementation.

SAY “The policy decides where to move; the PD controller computes the torque needed to move there.”

A transparent Gymnasium joint environment

TRANSPARENT DYNAMICS

$I \cdot \text{angular acceleration} = \tau - b \cdot \omega$

Semi-implicit Euler, $\Delta t = 0.05$ s:

$\omega_{next} = \omega + \Delta t(\tau - b\omega) / I$

$q_{next} = q + \Delta t \cdot \omega_{next}$

GYMNASIUM CONTRACT

5 continuous observations

1 bounded angle-target action

100 steps = 5 simulated seconds

Safety: $|q| > 2$ rad or $|\omega| > 12$ rad/s

```
reset(seed) → observation, info
```

```
step(action) → observation, reward,
terminated, truncated, info
```

SAY “The model is small enough that we can inspect every observation, force balance, and integration step.”

Reward encourages tracking; success is predefined

$$\text{reward} = 1 - 5e^2 - 0.005\omega^2 - 0.001\tau^2$$

TRACKING ERROR

Follow the desired angle.

ANGULAR VELOCITY

Discourage fast oscillations.

TORQUE

Limit excessive effort.

Success: full 5 s episode, no safety termination, RMSE < 0.32 rad

Threshold retained before training; 0.32 rad ≈ 18.3° is an educational tolerance.

SAY “The reward guides learning; the fixed RMSE threshold judges the final behavior.”

Matched PPO training conditions

MATCHED PPO RECIPE

SB3 PPO • MLP [32, 32] • CPU

12,288 timesteps per policy

Rollout 256 • batch 64 • 5 epochs

Learning rate 0.0008 • γ = 0.98

Same reward, references, seed 42

POLICY-GRADIENT CONNECTION

Sutton & Barto: general foundations

PPO: a specific actor–critic method with a clipped surrogate objective

Training-ratio clipping ≠ torque saturation

Experimental change: the distribution of training physics.

SAY “I changed the training physics distribution while keeping the other experimental choices consistent.”

Nominal, randomized, and held-out physics

Condition	Inertia scale	Friction b	Target delay
Fixed / nominal	1.00	0.20	0 ms
Randomized training	U(0.70, 1.80)	U(0.10, 0.65)	0, 50, 100, 150 ms
Held-out simulation	1.95	0.75	200 ms

Randomize at every reset; keep each episode’s physics constant.

Sensor SD: training / nominal = 0.005 rad, 0.02 rad/s

Held-out = 0.045 rad, 0.09 rad/s; noise affects policy and PD.

SAY “Randomization covers varied training joints; the held-out joint deliberately exceeds those ranges.”

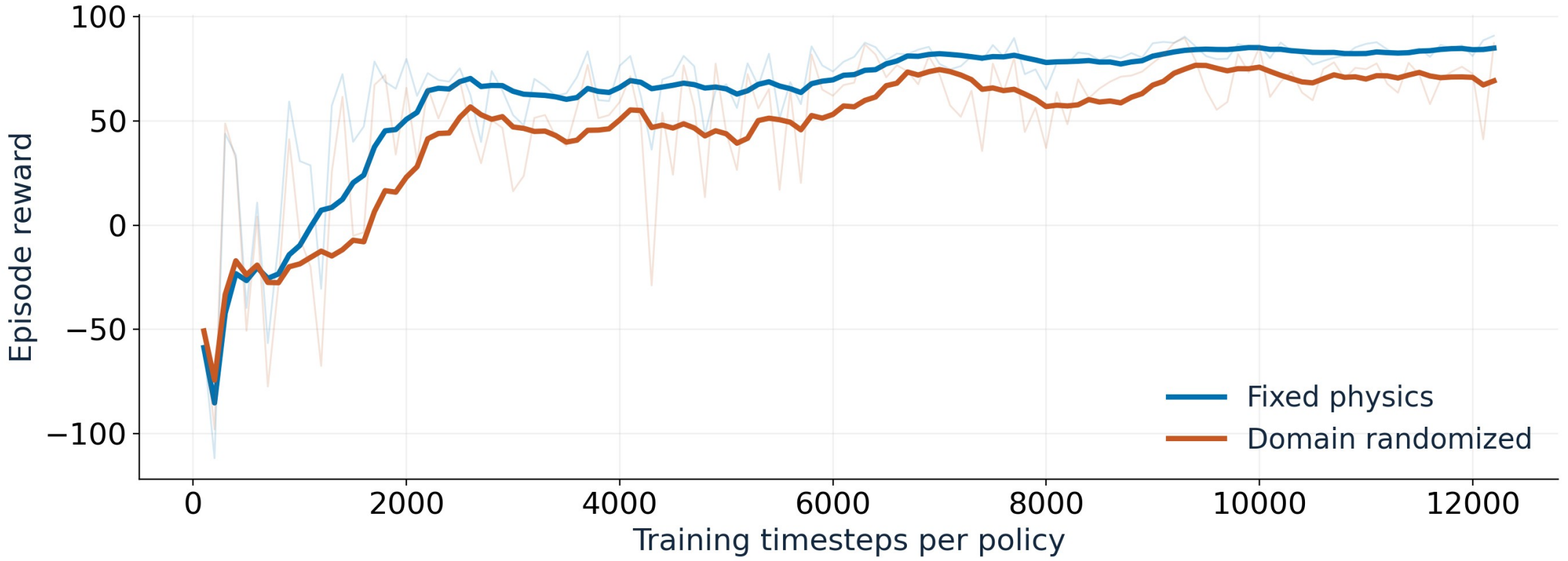
From readable code to auditable deliverables

Code or artifact	Purpose
Joint / check_env()	Physical model, API, and environment validation
train() + policy_*.zip	Two matched PPO training runs and saved controllers
roll() / summarize()	Paired evaluations and reproducible statistics
figures() + three PNGs	Recorded learning, tracking, and success
CSVs + tests + README	Audit trail, installation, and limitations

The classroom helper reuses saved models and writes into classroom/demo_runs/.

SAY “The files connect environment design, actual learning, evaluation, and evidence that another student can check.”

What the recorded training rewards show



Faint: actual episode returns • Bold: trailing 10-episode mean • Different training physics

SAY “These are actual training returns; the randomized policy is learning across a harder range of conditions.”

Evaluation: 30 matched trials per condition

30 MATCHED SEEDS

1000–1029

Same reference + initial state

Same underlying noise draws

120 EPISODES

2 policies × 2 environments

× 30 episodes

Deterministic evaluation

AUDITABLE METRICS

Reward and RMSE: mean ± SD

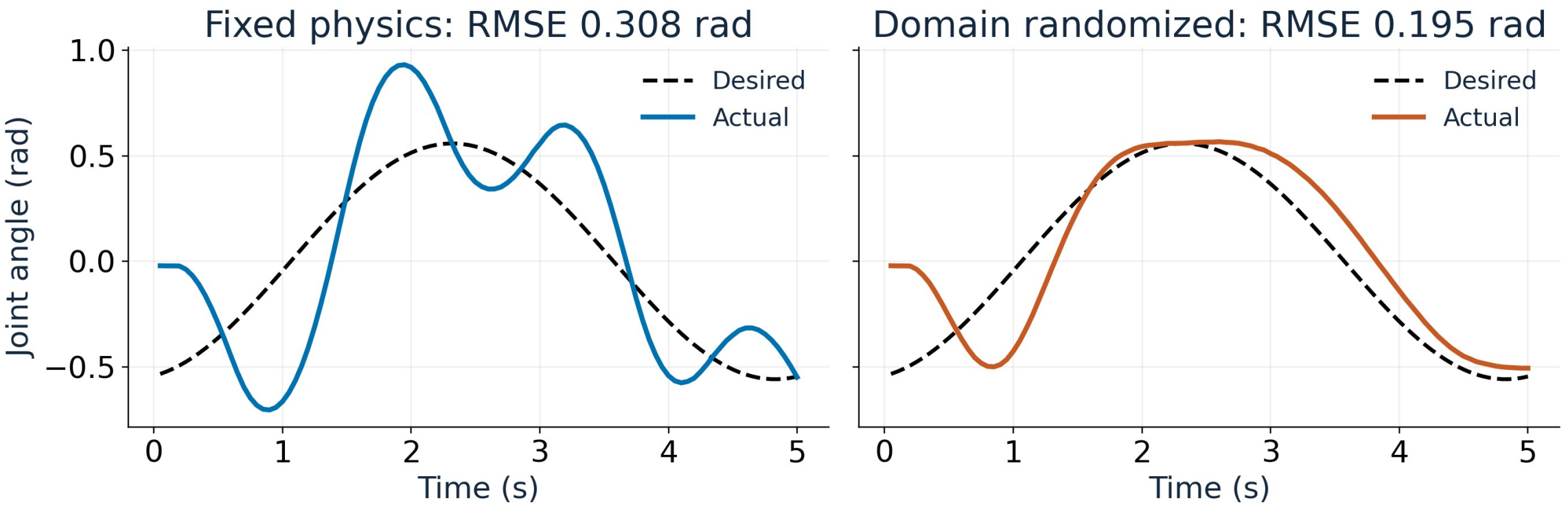
Success: percentage

95% Wilson intervals

The 30 evaluation seeds vary trials; there is only one training seed per policy.

SAY “The policies face matching trajectories and initial conditions, and every trial is saved.”

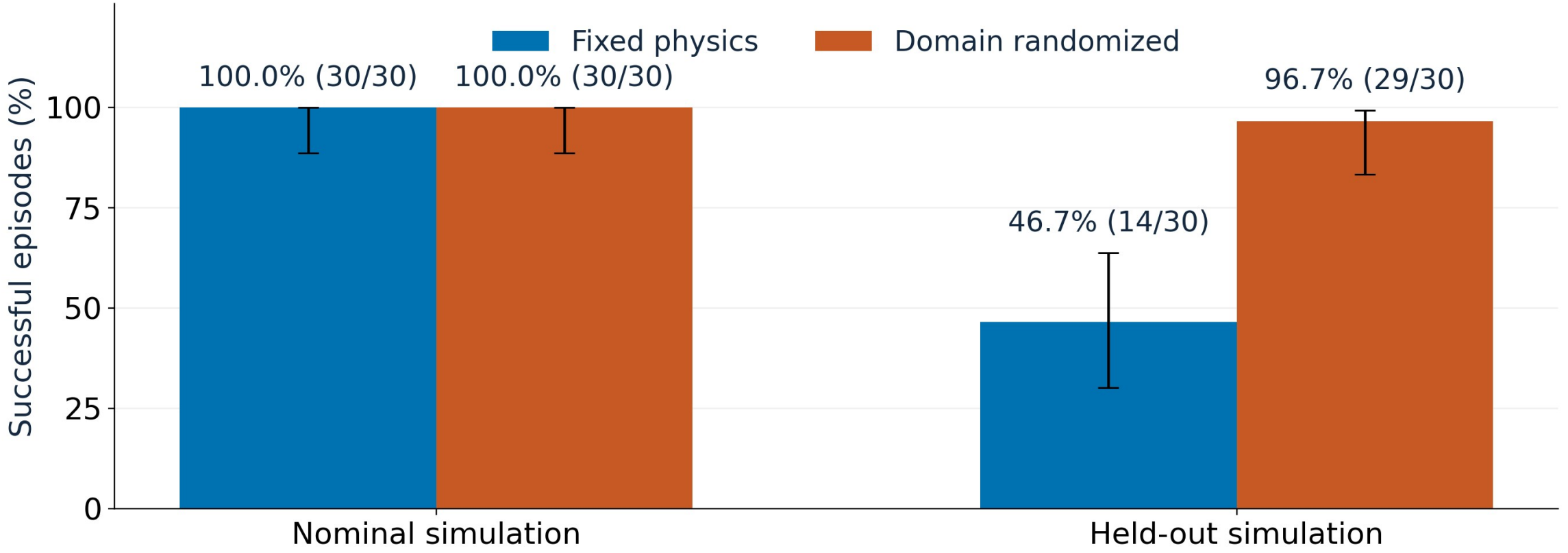
Held-out tracking: one matched example



Held-out seed 1000: both policies pass 0.32 rad. This one trial illustrates tracking, not the overall success rate.

SAY “This preselected trial shows tracking behavior; the full thirty-trial evaluation supports the overall comparison.”

Observed result: stronger held-out performance



Held-out mean episode RMSE: 0.3299 → 0.2250 rad • Intervals describe evaluation uncertainty

SAY “Domain randomization helped this trained policy pair in the held-out simulation; it is not a universal guarantee.”

Live demonstration: replay, compare, repeat

Replay the trained models; repeat the same matched seeds.

```
.\.venv\Scripts\python.exe classroom\demo.py --trials 3 --rounds 2
```

1 RUN

Load saved policies.

Watch both rounds finish.

2 COMPARE

Open the new tracking PNG.

Check identical rounds.

3 REPEAT

Run again: a fresh folder.

Or double-click run_demo.cmd.

Quick demo = 3 seeds; original results = 30. Repeated rounds add no independent evidence.

SAY “I am reloading trained policies, evaluating matching trials, and checking repeatable execution—not training again.”

Validation, provenance, and measured runtime

ENVIRONMENT

check_env(): all 3 modes

8 regression tests passed

Bounds, delay, dynamics

EXPERIMENT

7 artifact audits passed

12,000 evaluation steps

Saved-model replay

CLASSROOM

Two rounds match exactly

Full replay matches CSVs

Root results preserved

Measured cloud runtimes

Full experiment: 27.72 s • Training: 20.45 s

Quick replay: 5.82–7.06 s • Full replay: 9.32 s

Measured on the cloud CPU; laptop timing varies.

SAY “The plots are backed by raw records, executable checks, saved models, and measured wall-clock time.”

Limitations and the next engineering experiment

WHAT THIS RUN SUPPORTS

A controlled comparison of training physics

Observed robustness in one held-out condition

An inspectable mechatronics control loop

NEXT ENGINEERING TESTS

Multiple independent training seeds

Plain PD baseline and parameter ablations

Timestep sensitivity and richer dynamics

Calibrated hardware with safety systems

No full quadruped, contacts, balance, calibrated hardware, or proof of stability is claimed.

SAY “The next step is stronger evidence: more training seeds, classical baselines, and more realistic physics.”

Takeaway: robustness is an evidence-based claim

For this policy pair, varied physics improved held-out robustness.

14 → 29

Successful held-out trials / 30

0.3299 → 0.2250

Held-out mean RMSE (rad)

100%

Nominal success, both

Engineering takeaway: test the reality gap, retain the evidence, and qualify the claim.

SAY “Randomization improved this simulated transfer test; the data define how far that conclusion can go.”

How the four Reading 5 sources connect

TAN ET AL. (2018)

Sim-to-Real: Learning Agile Locomotion for Quadruped Robots

https://arxiv.org/abs/1804.10332 | Actuation and randomized-simulation motivation

ZHAO ET AL.

Sim-to-Real Transfer in Deep Reinforcement Learning for Robotics: A Survey

https://arxiv.org/abs/2009.13303 | Reality-gap context and transfer strategies

SUTTON & BARTO · CHAPTER 13

Reinforcement Learning: An Introduction

http://incompleteideas.net/book/the-book-2nd.html | Policy-gradient / actor–critic foundations

MUJOCO

Full multibody physics, contacts, constraints, and actuators

https://mujoco.org/ | A richer simulator; not used in this one-joint code

Specific PPO method: Schulman et al. (2017), https://arxiv.org/abs/1707.06347

SAY “The readings motivate the engineering choices; this small experiment does not reproduce their complete systems.”

Backup: complete verified numerical comparison

Policy	Condition	Reward ± SD	RMSE ± SD (rad)	Success	95% interval
Fixed	Nominal	93.42 ± 2.64	0.1022 ± 0.0225	100.0%	88.6–100.0%
Randomized	Nominal	93.67 ± 2.62	0.1037 ± 0.0235	100.0%	88.6–100.0%
Fixed	Held-out	34.08 ± 26.05	0.3299 ± 0.0734	46.7%	30.2–63.9%
Randomized	Held-out	70.97 ± 12.65	0.2250 ± 0.0505	96.7%	83.3–99.4%

Replay all 30 seeds without retraining:

```
.\.venv\Scripts\python.exe classroom\demo.py --trials 30 --rounds 1
```

SAY “These are the full thirty-trial statistics; quick replay values are labeled separately.”